National University of Lesotho
Department of Mathematics and Computer Science

CS4433 Software Engineering
Individual Final Project

Moderated Learning Management System

Requirements Engineering

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1. Introduction

This document covers requirements engineering for the Moderated Learning Management System (LMS). It has three parts, as the assignment brief asks: the stakeholder analysis (Section 2), the Software Requirements Specification or SRS (Section 3) and the use case model (Section 4). Section 5 gives the first version of the traceability matrix. Section 6 updates the risk register of the plan, because the plan says that the register is updated at every gate.

The statements of the stakeholders are incomplete and some of them conflict. Each sentence was read and put in one group: clear, unclear, conflict or missing. Each problem was settled once (Table 2) and the answer was written as a numbered requirement or business rule. The same numbers are used in the SRS, the use cases and the traceability matrix, so all the documents describe the same system. The numbers are: C for a problem in the statements, A for an assumption, FR for a functional requirement, NFR for a non-functional requirement, BR for a business rule, IF for an interface requirement and UC for a use case.

2. Stakeholder analysis

2.1 Stakeholders and their interests

Table 1 lists the six stakeholders of the system. Influence means how much the person can change what the system must do. Interest means how much the system affects the person's work. The last column shows which actor of the system the person is. The Principal and the IT officer do not use the system, so they are not actors. They are still stakeholders, because they set rules for it.

Table 1: Stakeholders, interests, influence and actor

Stakeholder	Role	Main interests	Influence	Interest	Actor in the system
Principal	Head of the school. Sponsor of the project.	The system must be simple for teachers who are not IT experts. Students must not see a mark before the Head of Department has checked it.	High	High	None. The Principal does not use the system.
Head of Department (HoD)	Checks the marks of the courses in the department.	Late work loses 10% of the available mark for each day late, for up to three days, then zero. Wants to see every mark in the department before it is released.	High	High	Head of Department
Lecturer	Teaches courses, sets assignments and marks work.	Students get their marks soon after marking. Can give a student more time when there is a good reason.	Medium	High	Lecturer
Students (the student representative spoke for them)	About 800 learners. Submit work and read marks.	Can fix a mistake and resubmit before the deadline. Written feedback, not only a number. Late work is not rejected.	Medium	High	Student

Stakeholder	Role	Main interests	Influence	Interest	Actor in the system
IT officer	Looks after the school's computers.	The system works offline. Passwords are protected. The system is quick with 800 students, on old computers.	Medium	Medium	None. The IT officer installs and supports the system.
System administrator	Looks after accounts and course set-up.	No statement was collected. Needs come from the case study: manage users, departments, courses, lecturers and enrolment.	Low	Medium	Administrator

The Principal and the Head of Department have high influence and high interest, so their statements win when there is a conflict. Lecturers and students have less power but use the system every day, so it must be easy for them. The IT officer sets the technical limits: offline use and old computers. The Administrator has low influence, but the system cannot work without the accounts and courses that the Administrator sets up.

2.2 Conflicts, unclear statements and missing information

Table 2 shows the 14 problems found in the statements and how each was settled. The same order of steps was used each time: (1) a clear rule of the sponsor (the Principal) or of the case study comes first; (2) if two statements only look different, a reading is used that makes both of them true; (3) if a statement cannot be tested, it is replaced by a number that can be tested; (4) if something is missing, the simplest choice that fits the case study is made and written down.

Table 2: Problems in the stakeholder statements and how they were settled

ID	What the stakeholders said	Problem found	What was decided and why	Result in
C1	Lecturer: "I want students to get their marks as soon as I have marked the work." Principal: "Students must not see any mark until it has been checked by the Head of Department." HoD: "I need to see every mark in my department before it is released."	Conflict. A mark cannot go to students straight away and also be checked first.	The Principal's rule is used, because the Principal is the sponsor and said it clearly. A marked submission is saved with the status Marked and students cannot see it. The HoD checks it and then releases it. To help the lecturer, the lecturer can see the status of every submission and gets a message when marks are released.	BR-02, BR-06, FR-19, FR-22, FR-23
C2	HoD: "Late work should lose 10% of the available mark for each day it is late, for up to three days. After that it should receive zero."	Unclear. "Available mark" could mean the maximum mark or the mark the student earned. "Day" could mean a calendar day or 24 hours.	"Available mark" means the maximum mark of the assignment. A "day" is each 24 hours that has started after the deadline. So the penalty is 10%, 20% or 30% of the maximum mark. Work more than 72 hours late gets zero. The final mark is never below zero. BR-01 gives the exact bands and the boundary times.	BR-01, FR-13
C3	HoD: "After that it should receive zero." Student representative: "Late work should not simply be rejected."	Looks like a conflict. A mark of zero may look like a rejection.	It is not a conflict. Late work is always accepted and saved. After 72 hours its mark is zero, but it is still saved and the lecturer still writes feedback. The student sees a warning about the penalty before the work is saved.	FR-18, BR-01

ID	What the stakeholders said	Problem found	What was decided and why	Result in
C4	Lecturer: "I should be able to give them an extension." HoD: a fixed penalty rule.	Missing. The statements do not say how an extension works with the penalty rule.	An extension is for one student and one assignment. It gives that student a new, later deadline. The penalty is counted from the new deadline. The lecturer must type a reason. The system saves who gave it and when. If the student had already submitted late, the days late are worked out again with the new deadline (BR-04).	BR-04, FR-10
C5	Student representative: "Students should be able to fix a mistake and resubmit before the deadline."	Unclear. It does not say which version is marked, if old versions are kept, or what happens after the deadline.	A student can resubmit until the student's deadline and until the work is marked. The latest version is marked. Older versions are kept. Late work cannot be replaced (BR-03).	BR-03, FR-17
C6	HoD: "I need to see every mark in my department before it is released."	Unclear. It does not say if the HoD only looks at marks or can also change them.	The HoD can approve a mark, change the final mark with a written reason, or send the work back to the lecturer with a comment. Every change goes into the audit log. A HoD works only on the courses of their own department.	BR-05, BR-07, FR-14, FR-21
C7	Principal: "the system must be simple." IT officer: "respond quickly even with all 800 students on it."	Unclear. "Simple" and "quickly" cannot be tested.	Each word becomes a number that can be tested. A main task needs at most 7 steps. Login takes at most 2 seconds with 800 students stored. Saving a record or showing a list takes at most 2 seconds.	NFR-01, NFR-02, NFR-06, NFR-07
C8	IT officer: "Passwords must be protected."	Unclear. "Protected" is not explained.	Passwords are saved only as salted hashes made with Argon2id (the same choice as in the project management plan). A password has at least 8 characters, with a letter and a digit. An account is locked for 5 minutes after 5 wrong passwords in a row.	NFR-03, BR-08, FR-03
C9	IT officer: "the system must work offline." Case study: "a single school computer or local network."	Unclear. It is not clear if several computers must share the data.	The system never needs the internet. Version 1.0 is installed on one school computer, as in the plan. Sharing the database between computers is not supported. Messages are shown inside the system and are not sent by email.	NFR-05, FR-23
C10	Student representative: "we want written feedback, not just a number." Principal: no mark may be seen before it is checked.	Missing. It is not said if feedback is compulsory, or if students may see feedback before the mark is released. Feedback could show the mark.	Written feedback is compulsory for every mark. Feedback stays hidden and is released together with the mark.	FR-12, FR-19, BR-02
C11	Case study: "submitted, late, marked, moderated, released."	Unclear. It is not said if "late" is a status or a fact about a submission.	A submission starts with the status Submitted or Late, depending on the time it arrives. The number of days late is saved separately, so it is kept when the status changes.	BR-06, FR-24
C12	Administrator: no statement was collected.	Missing. One of the four user types has no statement.	The Administrator's needs come from the case study: accounts, departments, courses, lecturers and enrolment. They are written as requirements and are confirmed with the supervisor at Gate 1.	FR-04 to FR-08

ID	What the stakeholders said	Problem found	What was decided and why	Result in
C13	Case study: "All changes to marks must be recorded so that the school can see who changed a mark."	Missing. It does not say who may look at the record.	Only the HoD can view the audit log, and only for the courses of the HoD's own department. The Administrator cannot view it, because it contains marks and the Administrator is not allowed to see marks.	FR-26, BR-10
C14	Principal: "Students must not see any mark until it has been checked by the Head of Department."	Missing. A status such as "Marked" could tell a student that marking is finished.	Until release, a student sees only the status Submitted or Late, whatever the real status is. After release the student sees Released with the mark and feedback.	FR-15, BR-02

2.3 Assumptions

Some facts that the requirements need are not in the case study. The assumptions below were made. They are confirmed with the supervisor at Gate 1.

- A1- The system is sized for 800 students and about 100 staff, so at most 1,000 user accounts. The case study gives only the number of students.
- A2- Version 1.0 is installed on one school computer, as in the project management plan. One person uses it at a time. Sharing one database between computers is not needed (see C9).
- A3- A submission is one file of at most 5 MB. The case study does not say what students hand in.
- A4- All times come from the clock of the school computer, in local time, and are recorded to the minute. A day is 24 hours.
- A5- Only the Administrator creates accounts. Nobody can register alone. The first Administrator account is made when the system is installed.
- A6- The reference computer used for the speed targets has a 2 GHz dual-core processor, 2 GB of memory and a hard disk drive. It stands for the school's old computers.
- A7- A course belongs to one department. Each department has one Head of Department, and a Head of Department leads one department. A lecturer can teach any course the Administrator assigns.
- A8- There is no last date for late work. Work that is more than 72 hours late is still accepted and gets zero.
- A9- A student who has not submitted has no mark. A lecturer cannot mark work that was not submitted.

3. Software Requirements Specification

This SRS follows the structure of ISO/IEC/IEEE 29148 (ISO/IEC/IEEE, 2018), made smaller to fit this project. Every requirement has a unique number, a MoSCoW priority, a source and a way to check it. The word "shall" is used in every requirement.

3.1 Introduction

3.1.1 Purpose

This SRS says what the Moderated LMS must do and how well it must do it. It is written for the supervisor, who marks it, and for the developer and tester, who build and check the system from it.

3.1.2 Scope

The system helps a school manage courses, assignments and marking. The Administrator manages users, departments, courses, lecturers and enrolment. Lecturers create assignments, give extensions, mark work and write feedback. Heads of Department check the marks of their department and release them. Students submit and resubmit work and see their marks and feedback after release. Every change of a final mark is recorded. The system is a C++ program that works offline on one school computer. The brief puts these things outside the scope: online payment, video conferencing, automated grading, plagiarism detection, and web or mobile interfaces.

3.1.3 Definitions

Table 3: Definitions

Term	Meaning
Administrator	The user who manages accounts, departments, courses, lecturers and enrolment.
Assignment	Work set by a lecturer for a course, with a deadline and a maximum mark.
Audit log	The permanent record of every time a final mark is set or changed.
Final mark	The raw mark minus the late penalty. It is never below 0.
Head of Department (HoD)	The user who checks (moderates) and releases the marks of one department.
LMS	Learning Management System.
MoSCoW	A way to rank requirements as Must, Should, Could or Won't have (Clegg and Barker, 1994).
Moderation	The check of marks by the Head of Department before they are released.
Raw mark	The mark that the lecturer types, before any late penalty.
Reference computer	The computer used for the speed targets (assumption A6).
Status	The stage of a submission: Submitted, Late, Marked, Returned, Moderated or Released.
Student's deadline	The assignment deadline, or the later deadline if the student has an extension.
Submission	The work of one student for one assignment. It can have more than one version.

3.2 Overall description

3.2.1 Product perspective

The system is a stand-alone program. It is not connected to any other school system or to the internet. It saves its data in one SQLite file, and all access to the file goes through a data-access layer.

3.2.2 Product functions

The main functions are: managing accounts and courses; creating assignments; submitting and resubmitting work; marking with an automatic late penalty; checking marks by the Head of Department; releasing marks; and recording changes of final marks in an audit log. Section 4 shows these functions as use cases.

3.2.3 User classes

There are four user classes. Administrators are school staff who do office work. Lecturers are teachers who are not IT experts. Heads of Department are senior teachers who check marks in groups. Students are about 800 learners, many under 18, who mainly submit work and read marks. Every user logs in with a username and a password and can use only the functions of the user's role (Table 10).

3.2.4 Operating environment and constraints

The system runs on the school's old computers with no internet. It is written in C++17, built with CMake and saves data in SQLite through a data-access layer that uses parameterised queries. Passwords are saved only as salted hashes. The project uses the Waterfall model with V-Model testing, so the requirements are fixed at Gate 1 and any later change must be written in the change log.

3.2.5 Assumptions and dependencies

The assumptions are in Section 2.3 (A1 to A9). The system depends on the SQLite and libsodium libraries and on the clock of the school computer, which gives every deadline and submission time.

3.3 External interface requirements

Table 4: External interface requirements

ID	Type	Requirement
IF-01	User interface	The system shall have a text menu interface that is used with the keyboard. A graphical interface is not needed.
IF-02	Hardware	The system shall run on a normal personal computer with a keyboard, a screen and a hard disk. It shall need no other device.
IF-03	Software	The system shall be written in C++17, built with CMake and shall save its data in one SQLite database file. It shall use the libsodium library for password hashes. It shall need no other software.
IF-04	Communication	The system shall not use any network or internet connection.

3.4 Functional requirements

Table 5 lists the functional requirements. The priority uses MoSCoW. Must means the system is not accepted without it. Should means it is important, but the system still works without it. Could means it is done only if there is time. The source shows who asked for the requirement. "Case study" means it comes from the text of the case study and not from a statement. Every functional requirement is tested by a system test or an acceptance test made from its use case (Phase 5).

Table 5: Functional requirements

ID	Requirement	MoSCoW	Source	Use case
FR-01	The system shall require every user to log in with a username and password before any other function can be used.	Must	Case study; IT officer	UC-01
FR-02	The system shall let each user use only the functions that Table 10 gives to the user's role. The system shall refuse any other action and show a message.	Must	Case study	All use cases
FR-03	The system shall lock an account for 5 minutes after 5 wrong passwords in a row.	Should	IT officer	UC-01
FR-04	The system shall allow the Administrator to create, edit and deactivate user accounts. Each account shall have a unique username and one role. An account with the role Head of Department shall also have a department. The system shall not allow two active Heads of Department in the same department.	Must	Case study	UC-02
FR-05	The system shall allow the Administrator to set a new password for a user who forgot the password.	Should	Case study	UC-02
FR-06	The system shall allow the Administrator to create, edit and deactivate departments and courses. Each course shall belong to exactly one department.	Must	Case study	UC-03
FR-07	The system shall allow the Administrator to assign one or more lecturers to a course.	Must	Case study	UC-04
FR-08	The system shall allow the Administrator to enrol students in a course and to remove them from it.	Must	Case study	UC-05
FR-09	The system shall allow a lecturer to create an assignment for a course the lecturer teaches. The assignment shall have a title, a description, a deadline (date and time) and a maximum mark.	Must	Case study	UC-06
FR-10	The system shall allow a lecturer to give one student an extension for one assignment of a course the lecturer teaches. The lecturer shall enter a new deadline and a reason. The system shall follow BR-04 and shall save who gave the extension and when.	Must	Lecturer	UC-07
FR-11	The system shall allow a lecturer to see all submissions for the assignments of the courses the lecturer teaches. For each submission the system shall show the status, the submission time and the days late.	Must	Lecturer	UC-08
FR-12	The system shall allow a lecturer to enter a mark from 0 to the maximum mark and written feedback for a submission. The system shall not save a mark without feedback.	Must	Case study; Student rep.	UC-08, UC-09
FR-13	The system shall work out the late penalty and the final mark of a submission by the rules of BR-01. Before saving, the system shall show the lecturer the raw mark, the penalty and the final mark.	Must	HoD	UC-08, UC-10

ID	Requirement	MoSCoW	Source	Use case
FR-14	The system shall save an audit log entry every time a final mark is set or changed. The entry shall have the user, the date and time, the old final mark, the new final mark and the reason (BR-05).	Must	Case study	UC-08, UC-11, UC-13
FR-15	The system shall allow a student to see only the courses in which the student is enrolled. For each course the system shall show the assignments with the student's deadline, the maximum mark and the status of the student's own submission (BR-02).	Must	Case study	UC-15
FR-16	The system shall allow a student to submit one file of at most 5 MB for an assignment on or before the student's deadline. The system shall save the file with the date and time and show a receipt.	Must	Case study	UC-16
FR-17	The system shall allow a student to replace a submission with a new file by the rules of BR-03. The system shall keep the older versions and use the latest version for marking.	Must	Student rep.	UC-18
FR-18	The system shall accept a submission made after the student's deadline and give it the status Late. Before saving, the system shall show the student the days late and the penalty that will apply. The system shall not reject late work.	Must	Student rep.; HoD	UC-17
FR-19	The system shall allow a student to see the final mark and the feedback of a submission only after its status is Released. Before that, the system shall not show the student any mark or feedback.	Must	Principal	UC-19
FR-20	The system shall allow a Head of Department to see every submission with the status Marked in the courses of the department. For each one the system shall show the raw mark, the penalty, the final mark, the lecturer and the feedback.	Must	HoD	UC-12
FR-21	The system shall allow a Head of Department to do one of three things for a submission with the status Marked: approve it, change its final mark and type a reason, or send it back to the lecturer and type a comment.	Must	HoD; Principal	UC-12, UC-13
FR-22	The system shall allow a Head of Department to release submissions with the status Moderated, one at a time or all of one assignment. The system shall not release a submission that is not Moderated.	Must	Principal; HoD	UC-14
FR-23	The system shall show a message at the next log in to a lecturer whose submissions were released or sent back, and to a student whose marks were released.	Should	Lecturer	UC-12, UC-14
FR-24	The system shall keep one status for each submission: Submitted, Late, Marked, Returned, Moderated or Released. The system shall allow only the changes shown in Table 9.	Must	Case study	UC-08, UC-12, UC-14, UC-16
FR-25	The system shall show a Head of Department, for each assignment of the department, the number of submissions that are Marked, Moderated and Released.	Could	HoD	UC-12
FR-26	The system shall allow a Head of Department to view the audit log entries of the courses of the department.	Must	Case study	UC-20

3.5 Non-functional requirements

Table 6 lists the non-functional requirements. Each one has a number or a test with a clear pass or fail result. Words such as "fast" and "simple" are not used without a number.

Table 6: Non-functional requirements

ID	Type	Requirement	MoSCoW	How it is checked	Source
NFR-01	Speed	Logging in shall take at most 2 seconds in at least 29 of 30 timed attempts on the reference computer, with 800 student accounts stored.	Must	Timed test	IT officer
NFR-02	Speed	Saving a submission of 5 MB, saving a mark and showing a list of 800 submissions shall each take at most 2 seconds on the reference computer.	Must	Timed test	IT officer
NFR-03	Security	Passwords shall be stored only as salted hashes made with Argon2id. Each password shall have its own random salt. The settings shall make one password check take between 100 and 500 milliseconds on the reference computer.	Must	Look at the code and database; test	IT officer
NFR-04	Security	Every function shall be refused to the roles that Table 10 does not allow. A student shall not be able to see the data of another student. In the tests, no function shall be available to a role that does not own it.	Must	Test of every function with every role	Case study; Principal
NFR-05	Offline use	Every function shall work on a computer that is not connected to any network.	Must	Run all acceptance tests with the network switched off	IT officer
NFR-06	Ease of use	After log in, each of these tasks shall need at most 7 steps: submit an assignment, mark a submission, approve a mark, release marks. A step is one menu choice or one typed entry.	Must	Count the steps in the menus	Principal
NFR-07	Ease of use	Five people who are not IT specialists shall try one task each with only the user manual. A student task is to submit an assignment. A lecturer task is to mark a submission. At least 4 of the 5 shall finish within 5 minutes.	Should	User trial	Principal
NFR-08	Reliability	The program shall not stop with an error for any of 30 wrong inputs or 5 simulated database errors. A failed action shall leave the saved data unchanged.	Must	Test	Assignment brief
NFR-09	Integrity	The program shall have no way to edit or delete an audit log entry. In the tests, every change of a final mark shall have exactly one audit log entry.	Must	Test; look at the code	Case study

ID	Type	Requirement	MoSCoW	How it is checked	Source
NFR-10	Maintainability	Automated tests shall cover at least 70% of the lines of the business-logic layer. A coding standard shall be written down. At Gate 3 no high-severity finding of the static analyser shall be left unfixed.	Must	Coverage report; static analysis report	Assignment brief
NFR-11	Installation	A school technician who follows the installation guide shall install and start the system in at most 30 minutes.	Should	Timed trial of the guide	IT officer
NFR-12	Accessibility	The program shall show all information as text and not by colour alone. Every function shall work with the keyboard only.	Should	Look at every screen and menu	Assignment brief
NFR-13	Size	The system shall store at least 1,000 user accounts, 100 courses and 20,000 submissions and still meet NFR-01 and NFR-02.	Must	Test with generated data	IT officer
NFR-14	Privacy	The system shall store only this personal data: name, username, role, department (Heads of Department only) and enrolments, with the submissions and marks.	Should	Look at the database schema	Feasibility study (legal)

3.6 Business rules

The business rules say how the requirements work together. The use cases, the design and the tests refer to them by number.

Table 7: Business rules

ID	Rule	Description
BR-01	Late penalty	D is the student's deadline: the assignment deadline, or the later deadline if the student has an extension. S is the time of the student's latest submitted version. M is the maximum mark. R is the raw mark that the lecturer entered. Times are compared to the minute. If S is not later than D, there is no penalty and the days late n is 0. If S is later than D, n is the time from D to S divided by 24 hours, rounded up to a whole number. If n is 1, 2 or 3, the penalty is n x 10% of M, and the final mark is R minus the penalty. The final mark is never below 0. If n is 4 or more (more than 72 hours late), the final mark is 0. M is a whole number and R has at most one decimal place, so the final mark has at most one decimal place and no rounding is needed.
BR-02	Hiding marks	A student cannot see the mark or the feedback of a submission until its status is Released. Until then the student sees the status Submitted (or Late if it was late), whatever the real status is.
BR-03	Resubmission	A student can replace a submission only if the time now is not later than the student's deadline and the status is Submitted. The system saves the new file as the next version and keeps the older versions. The latest version is the one that is marked. A submission with the status Late cannot be replaced.
BR-04	Extension	An extension is for one student and one assignment. The new deadline must be later than the student's current deadline. A reason is required. A lecturer of the course can give an extension only if there is no submission yet, or the status is Submitted or Late. The same student can get more than one extension, each later than the one before. If the student already submitted late, the system works out n again with the new deadline. If S is not later than the new deadline, the status changes from Late to Submitted, and the student can resubmit by BR-03.

ID	Rule	Description
BR-05	Audit log	The audit log records final marks. Each entry has the user, the date and time, the old final mark, the new final mark and the reason. The first mark has no old final mark and the reason "initial marking". Entries are kept for ever and are never changed.
BR-06	Status changes	The status changes only as shown in Table 9. A lecturer can change a mark only while the status is Marked or Returned, and must type a reason. A HoD can change a final mark only while the status is Marked, as part of moderation. A mark with the status Released cannot be changed in version 1.0.
BR-07	Moderation	A HoD can moderate and release only the courses of the HoD's own department. A submission must be Moderated before it can be Released.
BR-08	Passwords	A password has at least 8 characters, with at least one letter and one digit. The system never shows a password and never saves one as plain text.
BR-09	Marks	A mark is a number from 0 to the maximum mark, with at most one decimal place. A maximum mark is a whole number from 1 to 100.
BR-10	Access	Table 10 shows who can use which data. The Administrator cannot see submissions, marks, feedback or the audit log.

Table 8 shows the bands of BR-01 with an example. In the example the maximum mark M is 50 and the raw mark R is 40. These boundary times are used in the tests. A submission exactly 24 hours after D is in band 1. A submission 24 hours and 1 minute after D is in band 2. A submission exactly 72 hours after D is in band 3. A submission 72 hours and 1 minute after D gets zero. If R is smaller than the penalty (for example R is 10 and the penalty is 15), the final mark is 0 and not a negative number.

Table 8: Late-penalty bands with an example (M = 50, R = 40)

Band	Time of the submission	Days late (n)	Penalty (marks)	Final mark
On time	D or earlier	0	None	40
Band 1	Up to 24 hours after D	1	5	35
Band 2	More than 24 and up to 48 hours after D	2	10	30
Band 3	More than 48 and up to 72 hours after D	3	15	25
Band 4	More than 72 hours after D	4 or more	The final mark is 0	0

3.7 Submission status

Each submission has one status. The status changes only in the ways shown in Table 9. A state machine diagram of the same life cycle is given in the design document.

Table 9: Submission status

Status	Meaning	Set by	Next status
Submitted	The work arrived on or before the student's deadline.	The system, when the student submits	Marked
Late	The work arrived after the student's deadline. The days late are saved.	The system, when the student submits	Marked; or Submitted if an extension makes it on time (BR-04)
Marked	The lecturer entered a mark and feedback. Students cannot see it.	Lecturer	Moderated or Returned

Status	Meaning	Set by	Next status
Returned	The HoD sent the work back to the lecturer with a comment.	HoD	Marked
Moderated	The HoD approved the mark or changed it. Students cannot see it.	HoD	Released
Released	The HoD released the mark. The student can see the mark and the feedback.	HoD	None. This is the last status.

3.8 Who can use which data

Table 10 gives the access rules. "Own courses" means the courses that the lecturer teaches. "Own department" means the department that the HoD leads. The Administrator cannot see any submission, mark, feedback or audit log entry.

Table 10: Access by role

Data or function	Administrator	Lecturer	HoD	Student
Create and manage users, departments, courses, lecturers and enrolment	Yes	No	No	No
Create assignments and give extensions	No	Yes (own courses)	No	No
Submit and resubmit work	No	No	No	Yes (own work)
See the submitted files and the status of submissions	No	Yes (own courses)	Yes (own department)	Own submissions only
Enter and change marks and feedback	No	Yes (own courses)	No	No
Approve, change or send back marks, and release marks	No	No	Yes (own department)	No
See marks and feedback before release	No	Yes (own courses)	Yes (own department)	No
See marks and feedback after release	No	Yes (own courses)	Yes (own department)	Yes (own marks)
View the audit log	No	No	Yes (own department)	No

3.9 Priorities

Table 11 counts the requirements by priority. The Must requirements are the scope that is fixed at Gate 1. If the schedule is in danger, Could requirements are dropped first and then Should requirements.

Table 11: Requirements by MoSCoW priority

Group	Must	Should	Could	Total
Functional (FR)	22	3	1	26
Non-functional (NFR)	10	4	0	14
Total	**32**	**7**	**1**	**40**

These are Won't have in version 1.0: online payment, video conferencing, automated grading, plagiarism detection, web and mobile interfaces, messages by email or text (there is no internet), sharing the database between computers, users registering themselves, and an extension for a whole class at once.

4. Use case model

4.1 Actors

The system has four actors and one general actor. Table 12 describes them. The Principal and the IT officer are not actors, because they do not use the system (Table 1). The general actor User is used for generalization: the four roles are special cases of User, so they all share the use case Log in.

Table 12: Actors

Actor	Description	Use cases
Administrator	Manages users, departments, courses, lecturers and enrolment.	UC-02, UC-03, UC-04, UC-05
Lecturer	Creates assignments, gives extensions, marks submissions and writes feedback.	UC-06, UC-07, UC-08
Head of Department	Checks marks in the department, releases marks and views the audit log.	UC-12, UC-13, UC-14, UC-20
Student	Sees courses and assignments, submits work and sees released marks and feedback.	UC-15, UC-16, UC-17, UC-18, UC-19
User (general actor)	The general actor. The four actors above are special cases of User. Every one of them logs in.	UC-01

4.2 Use case diagram

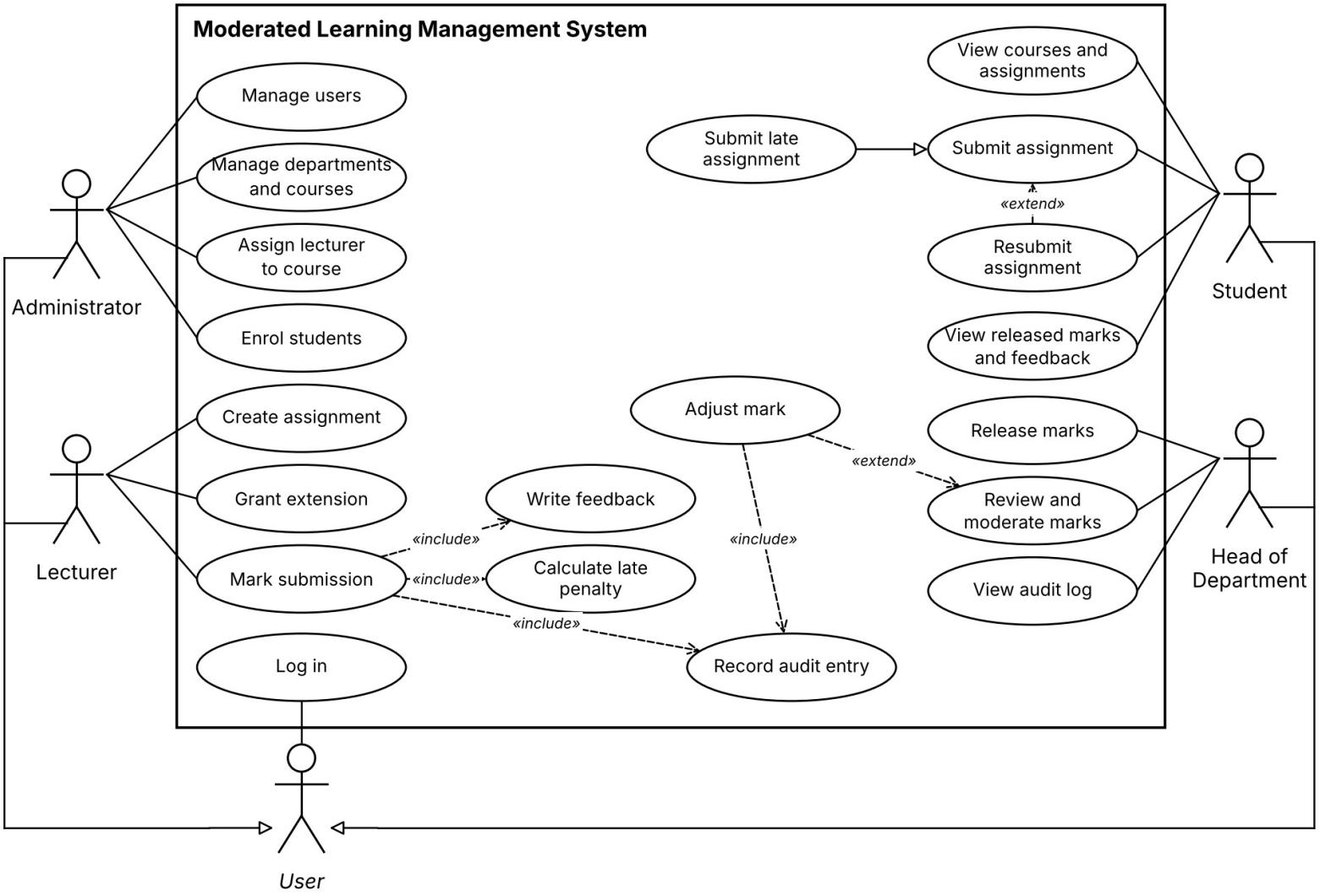


Figure 1: Use case diagram of the Moderated LMS

A dashed arrow is an include or extend relationship and points to the use case that is included or extended. A hollow triangle is generalization.

4.3 Use cases and relationships

The model has 20 use cases. Table 13 lists them with the requirements that each one meets. Figure 1 shows how they are connected. The relationships are:

- **Include.** Mark submission (UC-08) always uses Write feedback (UC-09), Calculate late penalty (UC-10) and Record audit entry (UC-11), because feedback and the penalty are compulsory for every mark and every final mark is recorded. Adjust mark (UC-13) also includes Record audit entry.
- **Extend.** Resubmit assignment (UC-18) extends Submit assignment (UC-16) only when the student already has a submission with the status Submitted and the deadline has not passed. Adjust mark (UC-13) extends Review and moderate marks (UC-12) only when the HoD does not agree with the final mark.
- **Generalization.** Submit late assignment (UC-17) is a special case of Submit assignment (UC-16), used when the time is after the student's deadline. The four actors are special cases of User.

Table 13: Use case list

ID	Use case	Actor	Purpose	Requirements
UC-01	Log in	User (all four roles)	Checks the username and password. Starts a session with the permissions of the role.	FR-01, FR-02, FR-03
UC-02	Manage users	Administrator	Creates, edits and deactivates accounts. Sets a new password.	FR-04, FR-05
UC-03	Manage departments and courses	Administrator	Creates, edits and deactivates departments and courses.	FR-06
UC-04	Assign lecturer to course	Administrator	Links lecturers to the courses they teach.	FR-07
UC-05	Enrolls students	Administrator	Enrolls students in a course or removes them.	FR-08
UC-06	Create assignment	Lecturer	Creates an assignment with a deadline and a maximum mark.	FR-09
UC-07	Grant extension	Lecturer	Gives one student a later deadline, with a reason.	FR-10
UC-08	Mark submission	Lecturer	Enters a mark and feedback for a submission.	FR-11, FR-12, FR-13, FR-14, FR-24
UC-09	Write feedback	Lecturer (included in UC-08)	Enters the written feedback that goes with a mark.	FR-12
UC-10	Calculate late penalty	System (included in UC-08)	Works out the penalty and the final mark by BR-01.	FR-13
UC-11	Record audit entry	System (included in UC-08 and UC-13)	Saves who set or changed a final mark, when, the old and new final mark and the reason.	FR-14
UC-12	Review and moderate marks	Head of Department	Checks marked work. Approves it or sends it back to the lecturer.	FR-20, FR-21, FR-23, FR-24, FR-25
UC-13	Adjust mark	Head of Department (extends UC-12)	Changes a final mark during moderation, with a reason.	FR-14, FR-21

ID	Use case	Actor	Purpose	Requirements
UC-14	Release marks	Head of Department	Makes moderated marks and feedback visible to students.	FR-22, FR-23, FR-24
UC-15	View courses and assignments	Student	Shows the student's courses, assignments, deadlines and status.	FR-15
UC-16	Submit assignment	Student	Saves the student's file and gives a receipt.	FR-16, FR-24
UC-17	Submit late assignment	Student (special case of UC-16)	Accepts work after the deadline, with a penalty warning.	FR-18
UC-18	Resubmit assignment	Student (extends UC-16)	Replaces an earlier submission before the deadline.	FR-17
UC-19	View released marks and feedback	Student	Shows the mark and feedback after release.	FR-19
UC-20	View audit log	Head of Department	Shows who changed final marks in the department, and when.	FR-26

4.4 Use case specifications

Three use cases are written in full (Cockburn, 2001): Submit assignment (UC-16), which the brief requires, Mark submission (UC-08) and Review and moderate marks (UC-12). Together they follow a mark from the student, to the lecturer, to the Head of Department. They cover the rules for late work, resubmission, the penalty, the audit log and moderation. The other use cases are described by the requirements in Table 5.

Table 14: Use case specification UC-16 Submit assignment

Use case	UC-16 Submit assignment
Primary actor	Student
Other actors	None
Description	A student submits one file as the answer to an assignment. This use case also covers a late submission (UC-17) and a resubmission (UC-18).
Related requirements	FR-15 to FR-19, FR-24, NFR-02, NFR-08
Trigger	The student chooses to submit work for an assignment.
Preconditions	1. The student has logged in (UC-01). 2. The student is enrolled in the course. 3. A lecturer has created the assignment.
Main flow (first submission, on time)	1. The student selects a course and then an assignment from the list shown by UC-15. 2. The system shows the assignment title, the maximum mark, the student's deadline and the status of any earlier submission. 3. The student chooses "Submit work" and types the path of the file. 4. The system checks that the file exists, is not empty and is not bigger than 5 MB. 5. The system compares the time now with the student's deadline (BR-01 and BR-04). 6. The time is not later than the deadline, so the system saves the file with the date and time as version 1 and sets the status to Submitted. 7. The system shows a receipt with the submission number, the date and time, and the status.

Use case	UC-16 Submit assignment
Alternative flows	A1. Late submission (UC-17). At step 5 the time is later than the deadline and not more than 72 hours later. The system shows the days late (n) and the penalty of n x 10% of the maximum mark. If the student confirms, the system goes to step 6, sets the status to Late and saves n. If the student cancels, nothing is saved and the use case ends. A2. More than 72 hours late. This is like A1, but the system shows that the mark will be zero. If the student confirms, the work is saved with the status Late. A3. Resubmission (UC-18). At step 2 the student already has a submission with the status Submitted and the deadline has not passed. The system asks the student to confirm that the new file replaces the old one. After the student confirms, the flow goes through steps 3 to 5. At step 6 the file is saved as the next version. The latest version is the one that is marked, and the older versions are kept.
Exception flows	E1. The file is not valid (step 4). The system shows the reason (file not found, file empty or file bigger than 5 MB) and goes back to step 3. Nothing is saved. E2. Resubmission is not allowed (A3). The deadline has passed, or the status is not Submitted. The system shows a message (BR-03) and the use case ends. E3. The student is not enrolled in the course, or the assignment does not exist (steps 1 and 2). The system refuses and the use case ends. E4. A database or disk error happens (step 6). The system cancels the save, leaves the earlier data unchanged and tells the student that the work was not saved.
Post conditions	Success: the file is saved as a new version with the date and time. The status is Submitted or Late. The days late are saved (0 if on time). The student has seen a receipt. Failure: no new version is saved and any earlier submission is unchanged.
Business rules	BR-01, BR-03, BR-04, BR-06

Table 15: Use case specification UC-08 Mark submission

Use case	UC-08 Mark submission
Primary actor	Lecturer
Other actors	None
Description	A lecturer gives a mark and written feedback for one submission. The system works out the late penalty and saves the final mark in the audit log.
Includes	UC-09 Write feedback, UC-10 Calculate late penalty, UC-11 Record audit entry
Related requirements	FR-11 to FR-14, FR-24, NFR-08, NFR-09
Trigger	The lecturer chooses to mark a submission.
Preconditions	1. The lecturer has logged in (UC-01). 2. The lecturer teaches the course. 3. The submission has the status Submitted, Late, Returned or Marked.
Main flow (first marking)	1. The lecturer selects a course, an assignment and then a submission from the list. 2. The system shows the submission, the student's name, the submission time, the student's deadline and the days late. 3. The lecturer types the raw mark, from 0 to the maximum mark. 4. The system checks the raw mark (BR-09). 5. UC-10: the system works out the late penalty by BR-01 and shows the raw mark, the penalty and the final mark. 6. UC-09: the lecturer types the written feedback. 7. The lecturer confirms. 8. The system saves the raw mark, the penalty, the final mark and the feedback, and sets the status to Marked. 9. UC-11: the system saves an audit log entry with the lecturer, the date and time, no old final mark, the new final mark and the reason "initial marking". 10. The system shows a confirmation. The student still cannot see the mark (BR-02).

Use case	UC-08 Mark submission
Alternative flows	A1. Returned submission. At step 2 the status is Returned. The system shows the HoD's comment. The lecturer follows steps 3 to 10 and also types a reason for the new mark. The audit log entry has the old final mark, the new final mark and this reason. A2. Change before moderation. At step 2 the status is Marked. The lecturer can type a new raw mark and a reason. Steps 4 to 10 follow and the status stays Marked. A3. Work more than 72 hours late. At step 5 the system shows that the final mark is 0 whatever the raw mark is.
Exception flows	E1. The raw mark is not a number or is outside 0 to the maximum mark (step 4). The system shows a message and goes back to step 3. E2. The feedback is empty, or the reason is empty in A1 or A2 (steps 6 and 7). The system does not save and asks for the missing text. E3. The lecturer does not teach the course. The system refuses and the use case ends. E4. The status is Moderated or Released (step 2). The system shows that the lecturer can no longer change the mark, and the use case ends. E5. A database error happens (step 8). The system cancels the save, leaves the earlier data unchanged and shows a message.
Post conditions	Success: the status is Marked. The raw mark, penalty, final mark and feedback are saved. One audit log entry has been saved. The student cannot see the mark. Failure: nothing is changed.
Business rules	BR-01, BR-02, BR-05, BR-06, BR-09

Table 16: Use case specification UC-12 Review and moderate marks

Use case	UC-12 Review and moderate marks
Primary actor	Head of Department
Other actors	None
Description	The Head of Department checks the marks of the courses in the department. For each submission the HoD approves the mark, changes the final mark, or sends the work back to the lecturer.
Extended by	UC-13 Adjust mark
Related requirements	FR-14, FR-20, FR-21, FR-23, FR-24, FR-25, NFR-02
Trigger	The HoD chooses to moderate marks.
Preconditions	1. The HoD has logged in (UC-01). 2. The HoD leads the department that owns the course. 3. At least one submission in a course of the department has the status Marked.
Main flow (approve a mark)	1. The HoD selects a course of the department. 2. The system lists the submissions with the status Marked. It also shows how many submissions of the assignment are Marked, Moderated and Released (FR-25). 3. The HoD selects a submission. 4. The system shows the student, the lecturer, the raw mark, the late penalty, the final mark and the feedback. 5. The HoD checks the mark and chooses "Approve". 6. The system sets the status to Moderated and saves who approved it and when in the history of the submission. 7. The HoD repeats steps 3 to 6 for other submissions, or goes on to release the marks (UC-14).
Alternative flows	A1. Change the final mark (UC-13). At step 5 the HoD chooses "Adjust" and types a new final mark from 0 to the maximum mark and a reason. The system saves the new final mark, saves an audit log entry (UC-11) with the old final mark, the new final mark and the reason, and sets the status to Moderated. A2. Send back to the lecturer. At step 5 the HoD chooses "Return" and types a comment. The system sets the status to Returned. The lecturer sees a message at the next log in and marks the work again (UC-08, A1).

Use case	UC-12 Review and moderate marks
Exception flows	E1. The HoD does not lead the department of the course. The system refuses and the use case ends (BR-07). E2. No submission is waiting (step 2). The system shows "No marked submissions to moderate" and the use case ends. E3. In A1 the new final mark is outside the range or the reason is empty, or in A2 the comment is empty. The system shows a message and goes back to the step. E4. A database error happens (step 6, A1 or A2). The system cancels the action, leaves the earlier data unchanged and shows a message.
Post conditions	Success: each submission handled is Moderated or Returned. Every changed final mark has an audit log entry. Students still cannot see any mark. Failure: nothing is changed.
Business rules	BR-02, BR-05, BR-06, BR-07, BR-09

5. Requirements traceability matrix (version 1)

Table 17 links each functional requirement to its use case. The columns for the design elements (classes, diagrams and tables) and the test cases are empty on purpose. They are filled in at Gate 2 and Gate 3. At the end, every requirement can then be followed from the stakeholder statement (Table 2), to the use case, the design, the code and the tests. The source of each requirement is in Table 5.

Table 17: Requirements traceability matrix, version 1 (Gate 1)

FR	Requirement	MoSCoW	Use case	Design element (Gate 2)	Test case (Gate 3)
FR-01	Log in	Must	UC-01		
FR-02	Who can do what	Must	All use cases		
FR-03	Account lock	Should	UC-01		
FR-04	Manage user accounts	Must	UC-02		
FR-05	Set new password	Should	UC-02		
FR-06	Departments and courses	Must	UC-03		
FR-07	Assign lecturers	Must	UC-04		
FR-08	Enrol students	Must	UC-05		
FR-09	Create assignment	Must	UC-06		
FR-10	Individual extension	Must	UC-07		
FR-11	View submissions	Must	UC-08		
FR-12	Mark and feedback	Must	UC-08, UC-09		
FR-13	Late penalty	Must	UC-08, UC-10		
FR-14	Audit log	Must	UC-08, UC-11, UC-13		
FR-15	View courses and assignments	Must	UC-15		
FR-16	Submit assignment	Must	UC-16		
FR-17	Resubmit	Must	UC-18		

FR	Requirement	MoSCoW	Use case	Design element (Gate 2)	Test case (Gate 3)
FR-18	Late submission	Must	UC-17		
FR-19	View released marks	Must	UC-19		
FR-20	HoD views marks	Must	UC-12		
FR-21	Moderate marks	Must	UC-12, UC-13		
FR-22	Release marks	Must	UC-14		
FR-23	Messages	Should	UC-12, UC-14		
FR-24	Submission status	Must	UC-08, UC-12, UC-14, UC-16		
FR-25	Assignment summary	Could	UC-12		
FR-26	View audit log	Must	UC-20		

6. Risk register update at Gate 1

The risk register in Table 7 of the project management plan was reviewed at Gate 1. The same scales were used (Table 6 of the plan: likelihood and impact from 1 to 3, and score = likelihood x impact). Risk R2 went down, one new risk (R11) was added and the other risks did not change. The review log of the plan (Table 8 of the plan) now reads: Gate 1 - R2 changed from 6 to 4, R11 added.

Table 18: Risk register changes at Gate 1

Risk	Description	Change at Gate 1	Reason and action
R2	Requirements are unclear or conflict, which causes rework (Project)	Likelihood 3 changed to 2. Impact stays 2. Score 6 changed to 4.	14 conflicts, unclear statements and gaps were found and settled (C1 to C14). The rules are written in Section 3.6.
R10	Extra features are added and the scope grows (Project)	No change. Score stays 4.	The MoSCoW priorities are now fixed (Table 11). A change needs an entry in the change log.
R8	Code, database and diagrams do not match each other (Project)	No change. Score stays 6.	Traceability matrix version 1 is started (Table 17). It is completed at Gates 2 and 3.
R11 (new)	The late-penalty and moderation rules have special cases that cause bugs. Examples: exactly 24 or 72 hours late, an extension given after a late submission, a mark sent back and marked again (Technical)	New. Likelihood 3, impact 2, score 6.	BR-01 and Table 8 give the boundary times. The penalty code is built and unit tested first, before the menus.

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